

490-HW1

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Exercise 1

```
# xbar = treatment group sample mean, cbar = control group sample mean
# xvar = treatment group sample variance, cvar = control group sample variance
hedge_g = function(n1,n2,xbar,xvar,cbar,cvar,study,alpha=0.1) {
  pooled_var = ((n1-1)*xvar+(n1-1)*cvar)/(n1+n2-2) # pooled variance for study
  g_hat = (xbar-cbar)/(sqrt(pooled_var)) # effect size estimate
  g_variance = (n1+n2)/(n1*n2) + (g_hat^2)/(2*(n1+n2-2))
  g_error = sqrt(g_variance) # standard error of effect size
  gtest = g_hat / g_error # test statistic
  pval = (1-pnorm(abs(gtest))) # 1-sided test
  UCL = g_hat + qnorm(1-alpha/2)*(g_error)
  LCL = g_hat - qnorm(1-alpha/2)*(g_error)
  results = cbind(study,g_hat,g_error,LCL,UCL,pval) # combine calculations
  return (results)
}
p1 = hedge_g(46,48,0.2316,0.2254,-0.0027,0.0007,154) # protocols
p2 = hedge_g(30,26,0.2811,0.1441,0.027,0.1139,156)
p3 = hedge_g(75,72,0.1894,0.1981,0.0443,0.4972,157)
p4 = hedge_g(12,12,0.093,0.1389,0.2277,0.0488,162)
p5 = hedge_g(32,34,0.1622,0.0961,0.0056,0.0955,163)
p6 = hedge_g(31,31,0.1837,0.1246,0.0943,0.1734,166)
p7 = hedge_g(27,27,0.6612,0.706,-0.0057,0.9891,303)
p8 = hedge_g(46,47,0.1366,0.1211,-0.0057,0.1291,306)
results = rbind(p1,p2,p3,p4,p5,p6,p7,p8)
results
```

##	study	g_hat	g_error	LCL	UCL	pval
## [1,]	154	0.7045465	0.2127680	0.35457432	1.0545187	0.0004642564
## [2,]	156	0.6826415	0.2758798	0.22885957	1.1364235	0.0066727072
## [3,]	157	0.2435844	0.1656104	-0.02882056	0.5159893	0.0706689006
## [4,]	162	-0.4396939	0.4135947	-1.11999660	0.2406088	0.1438675193
## [5,]	163	0.5140474	0.2504519	0.10209059	0.9260042	0.0200615634
## [6,]	166	0.2316031	0.2548787	-0.18763497	0.6508412	0.1817595610
## [7,]	303	0.7243994	0.2812824	0.26173097	1.1870677	0.0050070009
## [8,]	306	0.4045532	0.2095590	0.05985936	0.7492471	0.0267725787

Exercise 2

```
# d1, or d2: # of disease in vacc group , or in control group
# nd1, or nd2: # of no disease in vacc group, or in control group
options(scipen=10)
# output gives p-values in non-exponential form to facilitate interpretations
log_odds = function(study,d1,nd1,d2,nd2,alpha=.05) {
  w_hat = (d1*nd2)/(d2*nd1)
  w_star_hat = log(w_hat) # log-transformed effect size
  w_variance = 1/d1 + 1/nd1 + 1/d2+ 1/nd2
  w_error = sqrt(w_variance) # standard error of effect size
  w_ts = w_star_hat / w_error # test statistic
  pval = 2*(1-pnorm(abs(w_ts))) # 2-sided test
  UCL = w_star_hat + qnorm(1-alpha/2)*(w_error)
  LCL = w_star_hat - qnorm(1-alpha/2)*(w_error)
  results = cbind(study,w_star_hat,w_error,LCL,UCL,pval) # combine calculations
  return (results)
}
t1 = log_odds(1,4,119,11,128) # trials
t2 = log_odds(2,6,300,29,274)
t3 = log_odds(3,3,228,11,209)
t4 = log_odds(4,62,13536,248,12619)
t5 = log_odds(5,33,5036,47,5761)
t6 = log_odds(6,180,1361,372,1079)
t7 = log_odds(7,8,2537,10,619)
t8 = log_odds(8,505,87886,499,87892)
t9 = log_odds(9,29,7470,45,7232)
t10 = log_odds(10,17,1699,65,1600)
t11 = log_odds(11,186,50448,414,27197)
t12 = log_odds(12,5,2493,3,2338)
t13 = log_odds(13,27,16886,29,17825)
results = rbind(t1,t2,t3,t4,t5,t6,t7,t8,t9,t10,t11,t12,t13)
results
```

##	study	w_star_hat	w_error	LCL	UCL	pval
## [1,]	1	-0.93869414	0.59759932	-2.1099673	0.232579012	0.1162348967439
## [2,]	2	-1.66619073	0.45621529	-2.5603563	-0.772025185	0.0002600005157
## [3,]	3	-1.38629436	0.65834116	-2.6766193	-0.095969405	0.0352270680404
## [4,]	4	-1.45644355	0.14252864	-1.7357945	-1.177092550	0.0000000000000
## [5,]	5	-0.21914109	0.22792933	-0.6658744	0.227592183	0.3363294031696
## [6,]	6	-0.95812204	0.09952520	-1.1531878	-0.763056232	0.0000000000000
## [7,]	7	-1.63377584	0.47645532	-2.5676111	-0.699940564	0.0006057601788
## [8,]	8	0.01202060	0.06330057	-0.1120462	0.136087437	0.8493896782489
## [9,]	9	-0.47174604	0.23869881	-0.9395871	-0.003904956	0.0481181688525
## [10,]	10	-1.40121014	0.27463016	-1.9394754	-0.862944911	0.0000003357786
## [11,]	11	-1.41795573	0.08859116	-1.5915912	-1.244320241	0.0000000000000
## [12,]	12	0.44663468	0.73086399	-0.9858324	1.879101781	0.5411300305888
## [13,]	13	-0.01734187	0.26764737	-0.5419211	0.507237341	0.9483382271997

Exercise 3

```
adj_fisher = function(study,n,r,alpha=0.01) {
  z_hat = 1/2 * (log((1+r)/(1-r)))
  var_z = 1 / (n-3)
  error_z = sqrt(var_z)
  z_ts = z_hat / error_z # test statistic
  pval = 2*(1-pnorm(abs(z_ts))) # 2-sided test
  UCL = z_hat + qnorm(1-alpha/2)*(error_z)
  LCL = z_hat - qnorm(1-alpha/2)*(error_z)
  results = cbind(study,z_hat,error_z,LCL,UCL,pval) # combine calculations, return together
  return (results)
}

s1 = adj_fisher(1,10,0.68) # studies
s2 = adj_fisher(2,20,0.56)
s3 = adj_fisher(3,13,0.23)
s4 = adj_fisher(4,22,0.64)
s5 = adj_fisher(5,28,0.49)
s6 = adj_fisher(6,12,-0.04)
s7 = adj_fisher(7,12,0.49)
s8 = adj_fisher(8,36,0.33)
s9 = adj_fisher(9,19,0.58)
s10 = adj_fisher(10,12,0.18)
s11 = adj_fisher(11,36,-0.11)
s12 = adj_fisher(12,75,0.27)
s13 = adj_fisher(13,33,0.26)
s14 = adj_fisher(14,121,0.40)
s15 = adj_fisher(15,37,0.49)
s16 = adj_fisher(16,14,0.51)
s17 = adj_fisher(17,40,0.40)
s18 = adj_fisher(18,16,0.34)
s19 = adj_fisher(19,14,0.42)
s20 = adj_fisher(20,20,0.16)
results = rbind(s1,s2,s3,s4,s5,s6,s7,s8,s9,s10,s11,s12,s13,s14,s15,s16,s17,s18,
               s19,s20)
results
```

##	study	z_hat	error_z	LCL	UCL	pval
##	[1,]	1	0.82911404	0.37796447	-0.1444579270	1.8026860
##	[2,]	2	0.63283319	0.24253563	0.0081028165	1.2575636
##	[3,]	3	0.23418947	0.31622777	-0.5803592795	1.0487382
##	[4,]	4	0.75817374	0.22941573	0.1672379747	1.3491095
##	[5,]	5	0.53606034	0.20000000	0.0208944759	1.0512262
##	[6,]	6	-0.04002135	0.33333333	-0.8986311217	0.8185884
##	[7,]	7	0.53606034	0.33333333	-0.3225494312	1.3946701
##	[8,]	8	0.34282825	0.17407766	-0.1055660729	0.7912226
##	[9,]	9	0.66246271	0.25000000	0.0185053815	1.3064200
##	[10,]	10	0.18198269	0.33333333	-0.6766270792	1.0405925
##	[11,]	11	-0.11044692	0.17407766	-0.5588412431	0.3379474
##	[12,]	12	0.27686382	0.11785113	-0.0267005720	0.5804282
##	[13,]	13	0.26610841	0.18257419	-0.2041715311	0.7363883
##	[14,]	14	0.42364893	0.09205746	0.1865246225	0.6607732
##	[15,]	15	0.53606034	0.17149859	0.0943092555	0.9778114

```
## [16,] 16 0.56272977 0.30151134 -0.2139119874 1.3393715 0.061990525677
## [17,] 17 0.42364893 0.16439899 0.0001852012 0.8471127 0.009967468415
## [18,] 18 0.35409253 0.27735010 -0.3603139811 1.0684990 0.201708616077
## [19,] 19 0.44769202 0.30151134 -0.3289497332 1.2243338 0.137589799758
## [20,] 20 0.16138670 0.24253563 -0.4633436740 0.7861171 0.505785480498
```